PACKING LIST

Premium Alloys
12030 Windfern Rd
Houston, TX 77064
1-888-PALLOYS
713-300-9583

PACKING LIST	NUMBER 22815	PAGE 1	DATE 06/27/2025
SALES REP HOUSE ACCOUNT	FOB ORIGIN		CUST ID CACTUS WEL
SHIP VIA OTHER	FOB PYMNT Prepay		OUR S.O.# 21214
YOUR P.O.# TRP0050380			

APPROVED
By Heather Campbell at 4:19 pm, 6/27/25

SOLD TO:
Cactus Wellhead LLC
920 Memorial City Way
Suite 300
HOUSTON, TX 77024
Phone: (713) 626

SHIP TO:
Cactus Wellhead LLC
2123B MEADOW LANE
BOSSIER CITY, LA 71111

LINE	PCS	TOTAL LENGTH	ITEM ID/NAME	WIDTH	LENGTH	WEIGHT
1	8	18.333'	11OD718API 11 OD 718 API 6A CRA PER NACE MR0175 CMS-305 REV D, CNT-005 REV E LOT# 11		27.50000"	6,316

Warehouse	Tag #	Heat #	Mill #	PCS	Weight	Actual Length
MAIN	26351	LT7499EY11		8	6,316.000	27.5"

Phone: (713) 300-9583

TOTAL WT: 6,316
TERMS: NET 60

RECEIVED BY: _____

DATE: _____

CERTIFICATE OF CONFORMITY

No. 22815

Premium Alloys
12030 Windfern Rd
Houston, TX 77064
1-888-PALLOYS
713-300-9583

YOUR P.O.#		SHIP DATE
TRP0050380		06/27/2025
		OUR ORDER # 21214

SOLD TO
Cactus Wellhead LLC
920 Memorial City Way
Suite 300
HOUSTON, TX 77024
Phone: (713) 626

LINE	PCS	TOTAL LENGTH	DESCRIPTION	WIDTH	LENGTH	WEIGHT
1	8	18.333'	11 OD 718 API 6A CRA PER NACE MR0175 Item ID: 11OD718API CMS-305 REV D, CNT-005 REV E LOT# 11		27.50000"	6,316 LBS
			Warehouse / Tag # / Heat # / PCS / Length MAIN / 26351 / LT7499EY11 / 8 / 27.5000"			

* UNLESS OTHERWISE NOTED, THIS CERTIFIES THE ITEMS DESCRIBED ARE IN CONFORMANCE WITH THE REQUIREMENTS OF THE STATED PURCHASE ORDER AS EVIDENCED BY TEST REPORT OR OTHER RELATED DOCUMENTATION AND HAVE BEEN SUBJECT TO OUR CERTIFIED BS EN ISO 9001:2015 QUALITY MANAGEMENT SYSTEM

GEORGINA BIEHL

FOR AND ON BEHALF OF:
PREMIUM ALLOYS

TAG# 26351

HUNTINGTON ALLOYS CORPORATION
3200 Riverside Drive, Huntington, West Virginia 25705-1771 USA
Tel: +1.304.526.5100 Toll-Free in the USA: 1.800.334.4626
Fax: +1.304.526.5643 info@specialmetals.com

Certificate No. 148754-00
Dated 22-JUN-25
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CERTIFIED MATERIALS TEST REPORT

Note: The recording of false, fictitious or fraudulent statements or entries on this document may be punishable as a felony under federal statute.

This report relates only to the item(s) tested and may not be reproduced except in full.

Customer
PREMIUM ALLOYS INC
12030 WINDFERN RD.
HOUSTON
TX
77064

Ship To
PREMIUM ALLOYS INC
12030 WINDFERN RD.
HOUSTON
TX
77064

Sales Order Number	Purchase Order Number	Mark Order Number	Material Heat / Lot Identity
100083947 / 1.1	10061	10061	LT7499EY11

UNS Number
N07718

Material Description

INCONEL alloy 718, VACUUM INDUCTION MELTED-VACUUM ARC REMELTED, , HOT FIN ROUND, ROUGH T, SOL ANNEALED AND AGE HD, 11.0000, 72.0000, IN
1 PC PC# 1 6570 LBS

Specifications

SCHLUMBERGER 100237403 REV BB WITH 125 KSI MINIMUM YIELD STRENGTH / HARDNESS AT 39 HRC MAX / ASTM B637-23 CHEMISTRY ONLY / SMC HA 79 REV 10 (GRADE 120) / NACE MR0175(2021)-ISO 15156-3 (2020) / API 6ACRA (N07718) (120K) FIRST EDITION ERRATA 1, 2, & 3 ADDENDUM 1, 2, & 3 / API 6A 21ST EDITION ERRATA 1, 2, 3, 4, & 5 ADDENDUM 1, 2, & 3 / SMC QCP 285 REV 3.

ANALYSIS

	C %	MN %	FE %	S %	SI %	CU %	NI %	CR %	AL %
H	.014	.08	18.2	.0005	.07	.09	53.3	18.4	.55
Method	C/S	XRF	XRF	C/S	XRF	XRF	XRF	XRF	XRF

	TI %	MG %	CO %	MO %	NB %	TA %	B %	BI %	CA %
H	1.04	.0020	.12	2.89	5.17	.0064	.0035	<.000001	.0006
Method	XRF	AAS	XRF	XRF	XRF	XRF	OES	ICP-MS	OES

	P %	PB %	SE %	SN %
H	.005	<0.000001	<0.000001	.00055
Method	XRF	ICP-MS	ICP-MS	ICP-MS

	NB+TA
H	5.17

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CERTIFIED MATERIALS TEST REPORT

Method

ANALYSIS METHOD LEGEND

AAS - Atomic absorption
C/S - Carbon/Sulfur
ICP - Inductively Coupled Plasma
ICP-MS - Inductively Coupled Plasma Mass Spectrometry
OES - Optical Emission Spectroscopy
XRF - X-Ray Fluorescence

TENSILE TEST

ROOM TEMP TENSILE - LONG|LAB|MECHANICAL

PIECE	TEST TEMPER	LOCATION	TENSILE KSI	.2% YIELD KSI	EFF GA LNGTH IN	RED OF AREA%	ELONG%	ORIENT
1	SG	H	186.8	143.0	2.00	39.6	25.8	LONG

HARDNESS TEST

HARDNESS - 120 SURVEY|LAB|MECHANICAL

Piece 1
Test Temper SG
Location H
Hard Scale HRC

120 Survey Individual

Position	Surface	Mid Radius	Center
2 Indent 1	37.8	38.4	39.5
2 Indent 2	39.0	35.8	38.7
2 Indent 3	39.2	39.6	38.3
6 Indent 1	38.5	39.6	
6 Indent 2	38.5	38.6	
6 Indent 3	38.7	37.7	
10 Indent 1	37.0	36.5	
10 Indent 2	38.5	39.4	
10 Indent 3	38.9	39.9	

120 Survey Average

Position	Surface	Mid Radius	Center
2 Oclock AVG	38.7	37.9	38.8
6 Oclock AVG	38.6	38.6	

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CERTIFIED MATERIALS TEST REPORT

10 Oclock AVG	38.1	38.6

Piece 1
Test Temper SG
Location T
Hard Scale HRC

120 Survey Individual

Position	Surface	Mid Radius	Center
2 Indent 1	37.2	36.7	39.1
2 Indent 2	39.0	39.7	38.3
2 Indent 3	39.6	38.6	38.7
6 Indent 1	38.6	37.0	
6 Indent 2	38.9	39.6	
6 Indent 3	38.0	36.4	
10 Indent 1	36.0	36.4	
10 Indent 2	38.3	39.6	
10 Indent 3	38.6	38.1	

120 Survey Average

Position	Surface	Mid Radius	Center
2 Oclock AVG	38.6	38.3	38.7
6 Oclock AVG	38.5	37.7	
10 Oclock AVG	37.6	38.0	

HARDNESS - STANDARD|LAB|MECHANICAL

TEST TEMPER	POSITION	HRD SCALE	VALUE 1	VALUE 2	VALUE 3	HRD AVG
SG	SUR	HRC	38.9	39.2	38.5	38.9

HEAT TREATMENT

TREATMENT	FURNACE	TEMP 1 F	TEMP 1 HOLD	HOLD 1 UNITS	CHANGE 1	CHANGE RATE 1
SOL ANN	SF2	1885	2	HR	WQ	IN:77/OUT:86

TREATMENT	FURNACE	TEMP 1 F	TEMP 1 HOLD	CHANGE 1
AGE HRD	SF2	1455	6HR/30MIN	AC

REMARK 1

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CERTIFIED MATERIALS TEST REPORT

HEAT TREAT CHARTS ATTACHED

OTHER TESTS

ETCH TEST (ORDER REQUIRED)|LAB|ETCH LAB

PIECE ID	TEST	TEMPER	LOCATION	TEST VERDICT	REMARKS	ETCHANT DESCRIPTION
1	SG	H	P	TOTAL 2 ETCH/A604. 1 H&T		Aqua Regia

ETCH TEST TO ASTM A604|LAB|ETCH LAB

PIECE ID	TEST	TEMPER	LOCATION	ETCHANT DESCRIPTION	CLI RATING	CLII RATING	CLIII RATING	CLIV RATING
1	SG	H		Aqua Regia	A	A	A	A

GRAIN SIZE MEASUREMENT|LAB|METALLOGRAPHY

PIECE ID	TEST	TEMPER	POSITION	TEST ORIENT	AV GS ASTM	NRM or DUP	GS CONFIG
1	SG		SUR	LONGITUDINAL	4	DUPLEX	WIDE RANGE

GS ASTM RNG 1	GS ASTM RNG 2	LOCATION
2	8	H

PIECE ID	TEST	TEMPER	POSITION	TEST ORIENT	AV GS ASTM	NRM or DUP	GS CONFIG
1	SG		MR	LONGITUDINAL	3.5	DUPLEX	WIDE RANGE

GS ASTM RNG 1	GS ASTM RNG 2	LOCATION
2	7	H

PIECE ID	TEST	TEMPER	POSITION	TEST ORIENT	AV GS ASTM	NRM or DUP	GS CONFIG
1	SG		CEN	LONGITUDINAL	3	DUPLEX	WIDE RANGE

GS ASTM RNG 1	GS ASTM RNG 2	LOCATION
1	6	H

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CERTIFIED MATERIALS TEST REPORT

IMPACT CHARPY - TRANS|LAB|MECHANICAL

PIECE	TEST	TEMPER	LOCATION	SIZE	STRIKER RAD SIZE	NOTCH ORIENT	J/SQ CM	FT LB	FTLB AVG
1	SG		H	10x10mm	8 mm	C-L	77.8	45.9	43.8

JOULES	TEMP°F	LAT EXP IN	%SHEAR
62.2	-75	.025	20

PIECE	TEST	TEMPER	LOCATION	SIZE	STRIKER RAD SIZE	NOTCH ORIENT	J/SQ CM	FT LB	FTLB AVG
1	SG		H	10x10mm	8 mm	C-L	69	40.7	43.8

JOULES	TEMP°F	LAT EXP IN	%SHEAR
55.2	-75	.023	10

PIECE	TEST	TEMPER	LOCATION	SIZE	STRIKER RAD SIZE	NOTCH ORIENT	J/SQ CM	FT LB	FTLB AVG
1	SG		H	10x10mm	8 mm	C-L	76.1	44.9	43.8

JOULES	TEMP°F	LAT EXP IN	%SHEAR
60.9	-75	.021	10

MICROSTRUCTURE TEST|LAB|METALLOGRAPHY

PIECE	TEST	TEMPER	LOCATION	TEST VERDICT	AREA EVAL	POSITION	METHOD	ETCHANT DESCRIPTION
1	SG		H	P	SURFACE	SUR	MMP 6A 718	Seven Acids

MAGNIFICATION
500X

PIECE	TEST	TEMPER	LOCATION	TEST VERDICT	AREA EVAL	POSITION	METHOD	ETCHANT DESCRIPTION
1	SG		H	P	MID RADIUS	MR	MMP 6A 718	Seven Acids

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CERTIFIED MATERIALS TEST REPORT

MAGNIFICATION

500X

PIECE	TEST TEMPER	LOCATION	TEST VERDICT	AREA EVAL	POSITION	METHOD	ETCHANT DESCRIPTION
1	SG	H	P	CENTER	CEN	MMP 6A 718	Seven Acids

RATING	MAGNIFICATION
FIG A4	500X

LIST OF ATTACHMENTS

PHOTOS, HEAT TREAT CHARTS AND ULTRASONIC DATA SHEET ATTACHED

SMC IS IN COMPLIANCE WITH ROHS 1, ROHS 2, & ROHS 3.
THE FURNACES ARE CALIBRATED AND SURVEYED TO AMS2750G, MIL-STD-1684E, & API SEC6A.

LOCATION LEGEND: B = BACK C = CENTER F = FRONT H = HEAD M = MIDDLE T = TOE

TEST VERDICT LEGEND: P = PASS W = WAIVER
TEST TEMPER LEGEND
SG - Solution anneal + aged
CHEMISTRY IS REPORTED FROM REMELTED INGOT.
HEAT TREATMENT FURNACES ARE CALIBRATED TO AMS 2750, AS REQUIRED BY MIL-H-6875.

NO WELD REPAIR HAS BEEN PERFORMED ON THIS MATERIAL.
ACTUAL HOT WORKING REDUCTION RATIO: 4.7:1
REMELT INGOT SIZE: 24 IN

QUALIFICATION TEST COUPON WAS A PROLONGATION CUT FROM PRODUCTION MATERIAL AFTER HEAT TREATMENT.

FOR PRODUCT UP TO 12" DIAMETER, HEAT TREATMENT TIMES AND TEMPERATURES ARE NORMALLY MONITORED BY HEAT SINK THERMOCOUPLES, WITH LARGER DIAMETERS MONITORED BY CONTACT THERMOCOUPLES. ATTACHED FURNACE REPORTS DETAIL THE ACTUAL PRACTICE USED. WHEN HEAT SINK THERMOCOUPLES ARE USED, THE HEAT SINKS MEET API 6A REQUIREMENTS, BEING EQUAL TO OR

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UP TO 1/2" GREATER THAN THE LARGEST EQUIVALENT ROUND BAR WITHIN THE LOAD. FOR BAR DIAMETERS > 2" THE EMBEDDED THERMOCOUPLES SHALL BE >= 1" BELOW THE SURFACE OF THE HEAT SINKS. WHERE FURNACE LOADS CONTAIN BARS OF DIFFERENT DIAMETERS, THE MAXIMUM DIFFERENCE BETWEEN THE SMALLEST AND LARGEST BAR DIAMETER IS 2.0".

MATERIAL CAPABLE OF MEETING THE LIQUID PENETRANT REQUIREMENTS OF THE ORDER AFTER ADEQUATE SURFACE PREPARATION BY THE CUSTOMER TO PERFORM A VALID TEST.

ULTRASONICALLY TESTED AND ACCEPTABLE TO API 6A CLAUSES 10.4.2.12.2 AND 10.4.2.12.3 AND SATISFACTORY PER SMC QCP 285.

MICROSTRUCTURE SPECIMENS HAVE BEEN REMOVED FROM BAR PROLONGATIONS OBTAINED FROM EVERY HEAT TREAT LOT AFTER FINAL HEAT TREATMENT. A HEAT TREAT LOT CONSISTS OF BARS FROM THE SAME REMELT INGOT, OF THE SAME SIZE (EXCEPT LENGTH) AND PROCESSED IN THE SAME FURNACE LOADS FOR BOTH ANNEALING AND AGING.

MICROSTRUCTURE SPECIMENS AT NEAR SURFACE, MID-RADIUS, AND CENTER HAVE BEEN EXAMINED AT 100X AND 500X. NO TOPOLOGICAL DUPLEX GRAIN SIZE DISTRIBUTIONS PER ASTM E1181, I.E. BANDING, NECKLACE, OR CROSS-SECTION CONDITION, ARE PRESENT.
EXAMINATION FOR DELETERIOUS PHASES CONFIRMS FREEDOM FROM ACICULAR PHASES EXCEPT IN INDIVIDUAL, ISOLATED GRAINS NOT REPRESENTATIVE OF THE BULK MICROSTRUCTURE;
NO INDIVIDUAL GRAINS ARE SURROUNDED BY ACICULAR PHASES.
MICROSTRUCTURE IS ALSO FREE FROM CONTINUOUS NETWORKS OF SECONDARY PHASES ALONG GRAIN BOUNDARIES EXCEPT FOR INDIVIDUAL GRAINS OR ISOLATED FIELDS OF VIEW NOT REPRESENTATIVE OF THE BULK MICROSTRUCTURE.
THE MICROSTRUCTURE IS FREE FROM LAVES PHASE.
DISCRETE, ISOLATED PARTICLES OF SECONDARY PHASES OR CARBIDES ARE ACCEPTABLE.

TOTAL OF ALL CHEMICAL ELEMENTS NOT REPORTED IS CONSIDERED RESIDUAL, AND DOES NOT EXCEED. 0.50%

NIOBIUM (NB) EQUALS COLUMBIUM (CB).

THE MAXIMUM AND MINIMUM DIAMETERS IN THE HEAT TREAT BATCH ARE WITHIN 2 INCHES

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MACROETCH SATISFACTORY AS PERFORMED AND EVALUATED PER API 6A718 / API 6ACRA; ASTM A604 SEVERITY A OR BETTER.

SMC-HA WAS THE TESTING FACILITY

SMC-HA WAS THE MELTING FACILITY

SMC-HA WAS THE HEAT TREATMENT FACILITY

SMC-HA WAS THE FINAL HOT WORKING FACILITY

FOR ROD AND BAR PRODUCTS, THE MATERIAL SIZE WHEN HEAT TREATED MAY BE
LARGER THAN THE SIZE SHIPPED DUE TO SUBSEQUENT PROCESSING.

FOR ALL PRODUCTS 3/4 INCH DIAMETER AND OVER, TENSILE TEST SPECIMENS
WERE 0.500 INCH GAUGE DIAMETER X 2 INCH GAUGE LENGTH AS STANDARD
SPECIMENS, WITH THE AXIS OF THE SPECIMEN LOCATION WITHIN THE PARENT
MATERIAL AT THE CENTER FOR PRODUCTS 1.5 INCH OR LESS OR AT
MID-RADIUS FOR PRODUCTS OVER 1.5 INCH IN ACCORDANCE WITH ASTM E8.
CHARPY IMPACT SPECIMENS FOR PRODUCT GREATER THAN 1.5" ARE TAKEN FROM MID RADIUS LOCATION.
FOR 1.5" OR LESS CHARPY IMPACT SPECIMENS ARE TAKEN FROM PRODUCT CENTER.

FOR BAR PRODUCTS LESS THAN ¾ INCH DIAMETER, TENSILE TEST SPECIMENS WERE .350" DIAMETER
SPECIMEN WITH A 1.4" GAGE LENGTH OR 4D WITH THE AXIS OF THE SPECIMEN LOCATION WITHIN
THE PARENT MATERIAL AT THE CENTER.

ALL TEST RESULTS ARE REPORTED TO AT LEAST THE REQUIRED PRECISION BY THE ROUNDING METHOD OF ASTM E 29
UNLESS OTHERWISE REQUIRED BY PURCHASE ORDER OR SPECIFICATION.
COUNTRY OF ORIGIN: MELTED AND MANUFACTURED IN THE USA. DFARS PART 252.225.7009 AND 252.225.7008 COMPLIANT
THIS CERTIFICATION AFFIRMS THAT THE CONTENTS OF THIS REPORT ARE CORRECT AND ACCURATE AND THAT ALL TEST RESULTS AND
OPERATIONS PERFORMED BY SPECIAL METALS CORPORATION, INC. OR ITS SUBCONTRACTORS ARE IN COMPLIANCE WITH THE MATERIAL
SPECIFICATIONS.
QUALITY SYSTEM MEETS REQUIREMENTS OF DIRECTIVE 2014/68/EU (PRESSURE EQUIPMENT DIRECTIVE),
ANNEX 1, CHAPTER 4.3 PER ABS GROUP LTD CERTIFICATE 41734 (EXPIRES JANUARY 28, 2027)

HUNTINGTON ALLOYS CORPORATION IS AN ACCREDITED INDEPENDENT NADCAP
MATERIALS TESTING LABORATORY VIA CERTIFICATE NUMBER 3200230925 EXPIRATION DATE: NOVEMBER 30, 2025
FOR ALL TESTING SPECIFIED IN THE SCOPE OF ACCREDITATION.
MATERIAL PRODUCED UNDER QA SYSTEM DOCUMENTED IN HUNTINGTON ALLOYS CORP QA MANUAL REV. 56, DATED 01/25/2024

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CERTIFIED MATERIALS TEST REPORT

EN 10 204 / DIN 50049 (TYPE 3.1)
QUALITY SYSTEM CERTIFICATION: AS9100D:2016, EN9100:2018, JISQ9100:2016 and ISO9001:2015
(ABS-QE CERT 56476 EXPIRES 1/28/2027)
LABORATORY IS ACCREDITED TO ISO/IEC 17025:2017 FOR MECHANICAL TESTING AND CHEMICAL ANALYSIS.

VISUAL AND DIMENSIONAL EXAMINATION SATISFACTORY.
MATERIAL, WHEN SHIPPED, IS FREE FROM CONTAMINATION BY MERCURY, RADIUM, ALPHA SOURCE, AND LOW MELTING ELEMENTS.

CHEMICAL ANALYSIS AS REQUIRED FOR CARBON, SULFUR, NITROGEN, OR OXYGEN IS PERFORMED BY COMBUSTION TECHNIQUES.
ALL OTHER REPORTED ELEMENTS ARE ANALYZED BY X-RAY AND/OR EMISSION SPECTROSCOPY.""

GE Transportation Aviation Special Process Certification Source Code 47150, Expiration Date of Dec 30, 2025

AUTHORIZED QUALITY CERTIFICATION REPRESENTATIVES:
D.L. CONLEY K.R. BREWER C.S. GUNNOE G.J. BURKHEAD T.A. WILKS

End Of Certificate

This is to certify that all required samplings inspections and tests have been performed in accordance with the order and specification requirements. The test report represents the actual attributes of the material furnished and the values shown are correct and true. The material described by this certificate is in full compliance with all order and inspection requirements. We hereby certify that the figures given are in accordance with the specified contract requirements.

REV. 8/08

Signed
For and on behalf of HUNTINGTON ALLOYS CORPORATION
Authorized Signature

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EF595

SPECIAL METALS

Ultrasonic Test Report

H-2245 Rev. 9/2023

Job Number	R1839471-01	**QC Procedure**	QCP 285 Rev 3 Date 03-03-2014
Heat Lot Number	LT7499EY11	**Additional Instruction**	**Supplement** **Rev** **Date** **Addendum** **Other** **Other**
Nom Diameter or X-section, wall or pipe schedule (in)	11.000"		

Number of Pieces **Piece Numbers if applied** **Serial Numbers if applied**	1 1	**Date of Inspection YYYY-MM-DD**	2025-06-07	
Time of test, with respect to final length, width and heat-treat condition	Tested at final cross-section Tested prior to cutting final length / width Tested as final heat-treat condition	**Inspector and Certification Levels**	L. Perry SNT-TC-1ALevel II MIL-STD-2132 Inspector	

Total Number of Probes Used for Test	1	**Shear Wave**				**Longitudinal Wave**	
		Circumferential Propagation		**Axial Propagation**		**Flaw Detection**	**Wall Thickness** ☐
		#1	**#2 Opposing**	**#1**	**#2 Opposing**	Contact Method	**Back Echo Monitor** ☐
Instrument Manufacture Model Serial Number Linearity Date m-d-y Frequency Filter Setting Cal Gain / Scan Gain dB		/	/	/	/	Staveley-Sonic 237 142PB 05-09-2025 2.25 MHz 41.0 / 47.0	/
Transducer Manufacture Model Serial Number Frequency MHz Element Size (in) Focal Length (in) Effective Beam (in) Refracted Angle Cable length/Type (ft) Reject On / Off		/	/	/	/	Single Element Technisonic GAMMA RHP 13E026XD 2.25 1.0 dia. 0 7.5 /RG / 174/U OFF	/
Scanning Method Speed (ips) Index Type Max Index Used (in) Max Pulse Spacie (in) Couplant						Manual 6 Step Index Oil	
Reference Amplitudes % FSH % FSH Back Echo % Back Echo Loss EDAC Used						 80 80 50 Yes	
Gating Delay and Range 1 Logic 1 % Ref Alarm Level 1 Delay and Range 2 Logic 2 % Ref Alarm Level 2						 FS to BR Positive 50 BR Negative 50	
Reportable Conditions (Indications or BR loss)						None	
Remarks	1 WATER WASHABLE LP INSPECTION PERFORMED FOR UT SHEAR WAVE. 2. 3.						

Calibration Standard Data (all measurements in inches)

Calibration Standard		**NLOD Depth**	**NLID Depth**	**NTOD Depth**	**NTID Depth**	**FBH Diameter**	**Wall Thickness Reference 1**
SN	K356					0.251, 0.250, 0.250	
Standard Finish	No smoother than material	**NLOD Length**	**NLID Length**	**NTOD Length**	**NTID Length**	Metal Travel ☒ Drilled Depth ☐	**Wall Thickness Reference 2**
Material Finish	Less than 6.3 Micrometers					10.51, 5.658, 2.584	
Acoustic Comp Check	Long Wave Checked, 0 dB compensation required			Shear Wave ,		dB compensation required	

☐ **Minimum Wall Thickness Maintained** **Strip Chart Recorder Used** NA **Scanning Equipment**

☒ **Inspected Material Accepted** **Attachments**

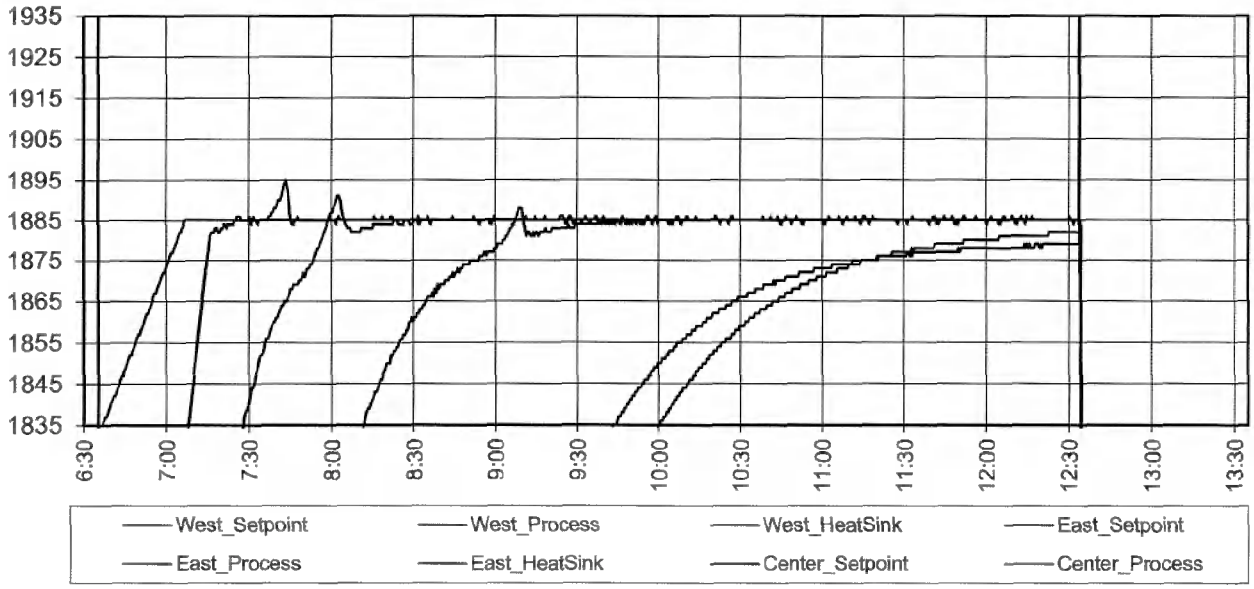
"NOTE: The recording of false, fictitious or fraudulent statements or entries on this document may be punishable as a felony under Federal Statute."

Order Report - Machine Shop Tip-Up Furnace #2 (50 Ft.)

Date & Time:	6/5/25	6:35:00
Operator:	13608	
Heat Type:	Solution Anneal	
Recipe #	167	
Quench:	Yes	
Water Temp (Start/End):		77 / 86
Maximum Water Temp:		86

Heat Sink Information	
East Heat Lot#	037101-25
East Heat Offset	4 °F
East heat Sink Size	11.250
West Heat Lot#	024852-25
West Heat Offset	3 °F
West Heat Sink Size	11.250

	Recipe Information			Event Information			
Event	Temp. SP	Hold Time (In Min.)	Ramp Rate (Per Hr.)	Start Date	Start Time	End Date	End Time
Charge				6/5/2025	6:35:00		
Step #1	1,885 °F	120	0 °F	6/5/2025	10:32:45	6/5/2025	12:32:45
Step #2	0 °F	0	0 °F				
Step #3	0 °F	0	0 °F				
Step #4	0 °F	0					
Discharge				6/5/2025	12:34:29		
Quench Time	60	Start Recipe HS Offset	25 °F	6/5/2025	12:34:52	6/5/2025	13:34:56



Order	Heat	Lot	Alloy	Weight	Dim1	Dim2	Dim3	Pieces	Location
R1835851-01	LT7500EY	11-1	718	7600	0	11	269.875	1	C11
R1839471-01	LT7499EY	11-1	718	7680	0	11	273.375	1	C12

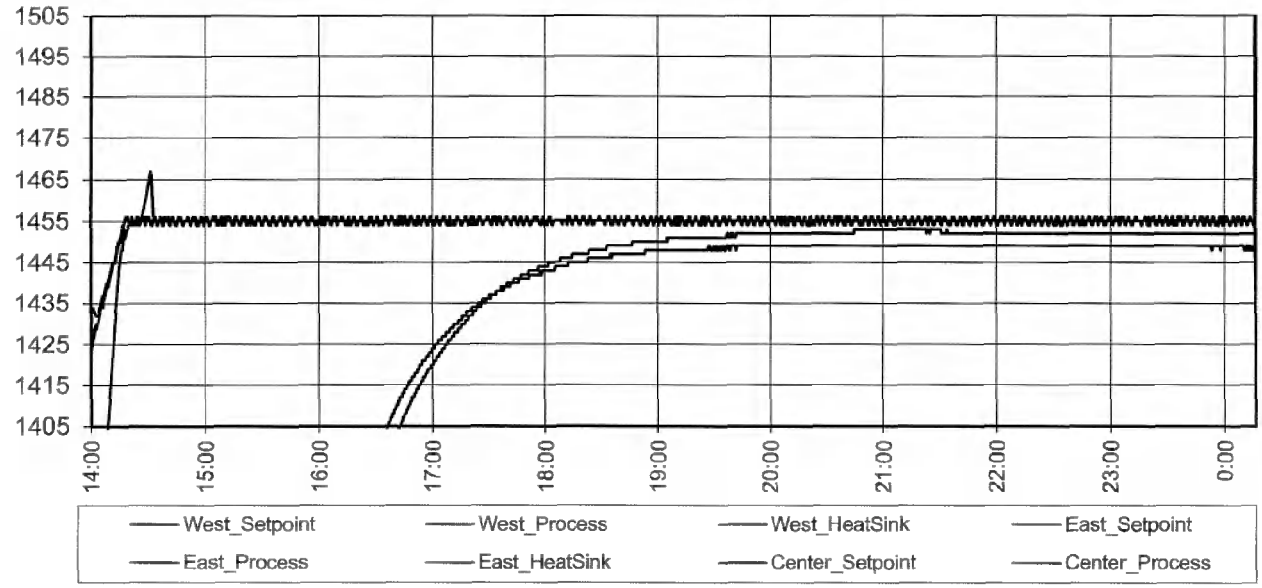
THE RECORDING OF FALSE, FICTITOUS OR FRAUDULENT STATEMENTS OR ENTRIES ON THIS DOCUMENT MAY BE PUNISHABLE AS A FELONY UNDER FEDERAL STATUTE.

Order Report - Machine Shop Tip-Up Furnace #2 (50 Ft.)

Date & Time:	6/5/25	13:46:45
Operator:	13314	
Heat Type:	Age Hard	
Recipe #	132	
Quench:	No	
Water Temp (Start/End):		

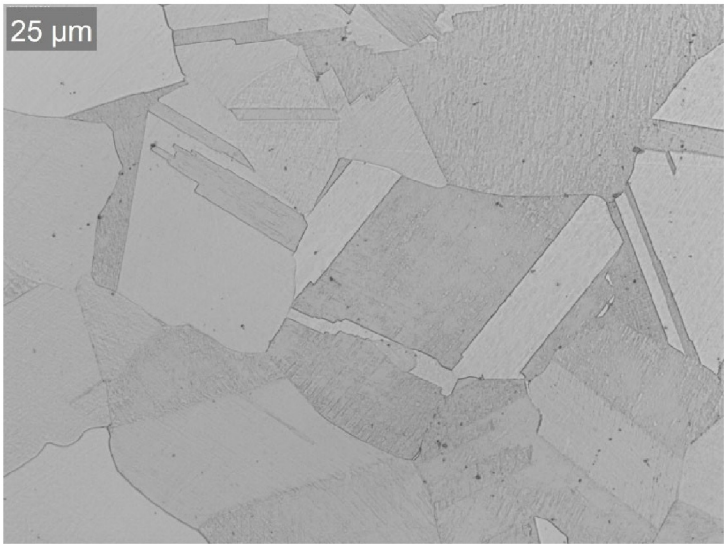
Heat Sink Information	
East Heat Lot#	037101-25
East Heat Offset	3 °F
East heat Sink Size	11.250
West Heat Lot#	024852-25
West Heat Offset	2 °F
West Heat Sink Size	11.250

Event	Recipe Information			Event Information			
	Temp. SP	Hold Time (In Min.)	Ramp Rate (Per Hr.)	Start Date	Start Time	End Date	End Time
Charge				6/5/2025	13:46:45		
Step #1	1,455 °F	390	0 °F	6/5/2025	17:44:37	6/6/2025	0:14:39
Step #2	0 °F	0	0 °F				
Step #3	0 °F	0	0 °F				
Step #4	0 °F	0					
Discharge				6/6/2025	0:16:23		
Quench Time		Start Recipe HS Offset	15 °F				



Order	Heat	Lot	Alloy	Weight	Dim1	Dim2	Dim3	Pieces	Location
R1835851-01	LT7500EY	11-1	718	7600	0	11	269.875	1	C11
R1839471-01	LT7499EY	11-1	718	7680	0	11	273.375	1	C12

THE RECORDING OF FALSE, FICTITOUS OR FRAUDULENT STATEMENTS OR ENTRIES ON THIS DOCUMENT MAY BE PUNISHABLE AS A FELONY UNDER FEDERAL STATUTE.



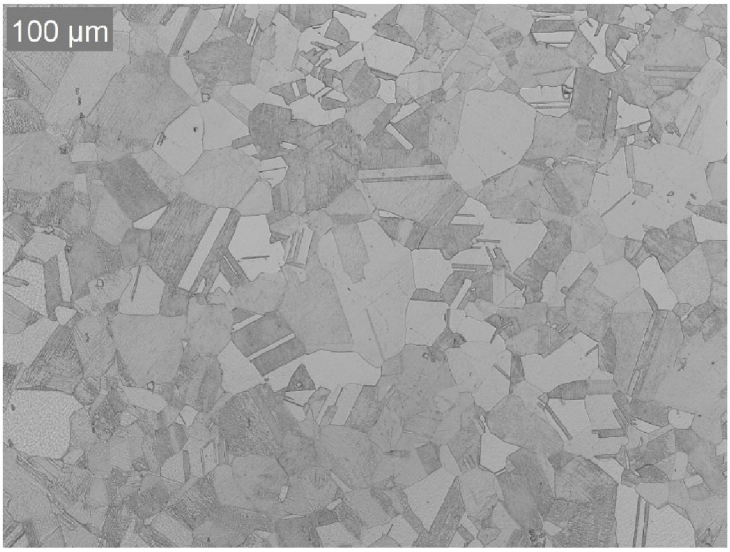


Figure 1: MS28346

Figure 2: GS28347

SPECIAL METALS

Sales Order Number	100083947-1
Heat Number/Lot No.	LT7499EY11
Piece I.D.	1 HEAD
Alloy	INCONEL alloy 718
Product Form	Round
Product Size	11.0000"
Product Condition	Solution Ann and Aged
Lab Heat Treat	None
Image Number	See Photomicrograph
Magnification	500x-100x
Orientation	Long-Sur
Etchant	7-Acids
Etch Time	Standard

NOTE: The recording of false, fictitious or fraudulent statements or entries on this document may be punishable as a felony under Federal Statute.

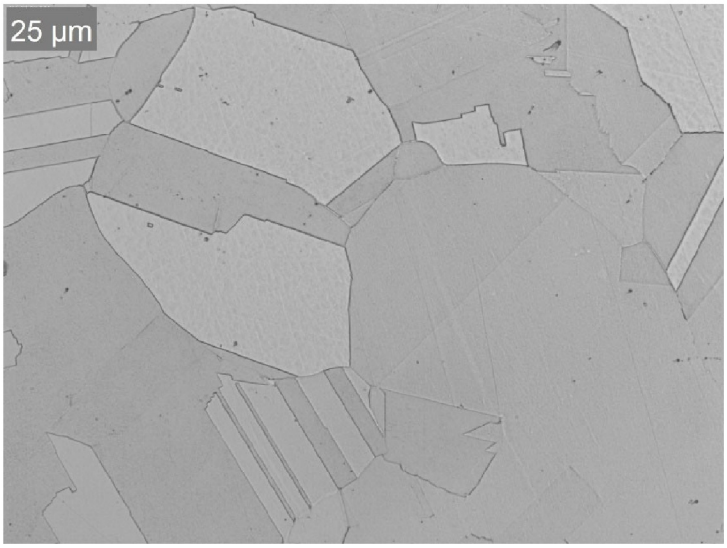


Figure 1: MS28348

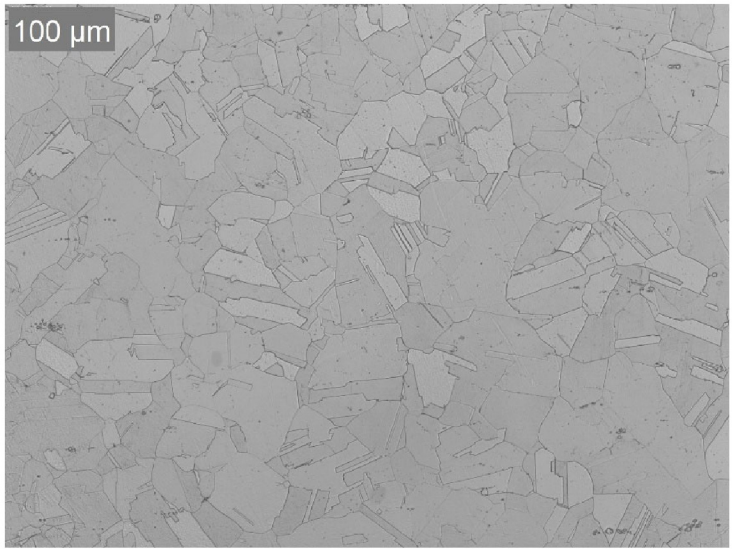


Figure 2: GS28349

SPECIAL METALS

Sales Order Number	100083947-1
Heat Number/Lot No.	LT7499EY11
Piece I.D.	1 HEAD
Alloy	INCONEL alloy 718
Product Form	Round
Product Size	11.0000"
Product Condition	Solution Ann and Aged
Lab Heat Treat	None
Image Number	See Photomicrograph
Magnification	500x-100x
Orientation	Long-Mr
Etchant	7-Acids
Etch Time	Standard

NOTE: The recording of false, fictitious or fraudulent statements or entries on this document may be punishable as a felony under Federal Statute.

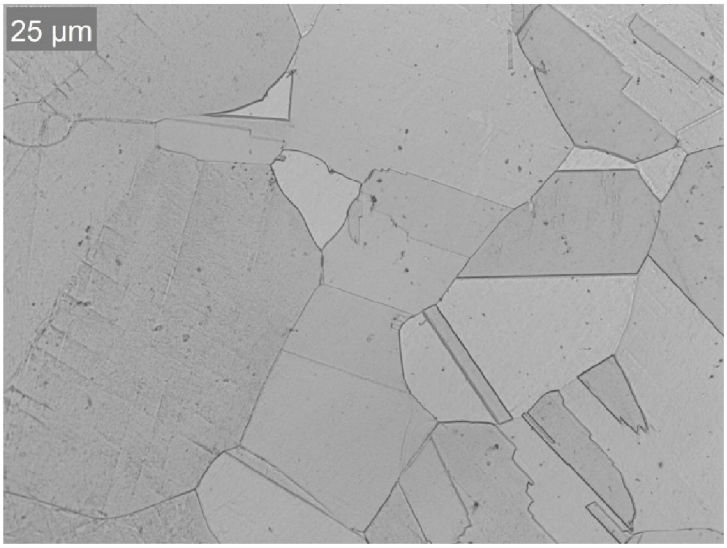


Figure 1: MS28350

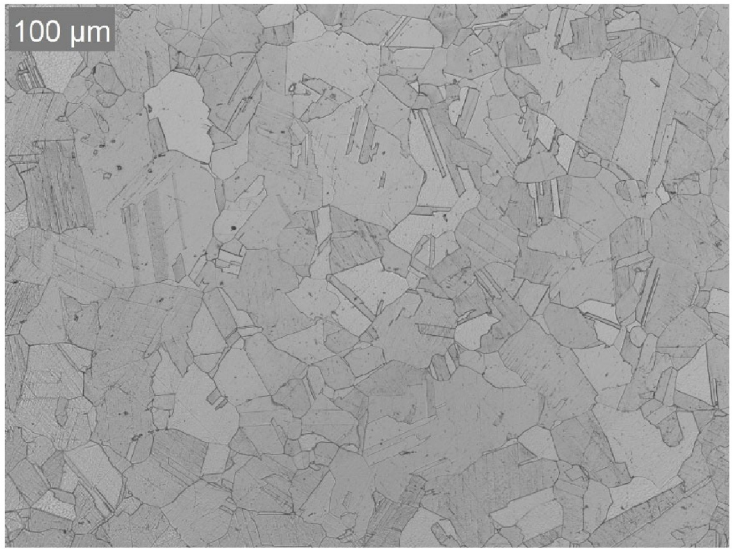


Figure 2: GS28351

SPECIAL METALS

Sales Order Number	100083947-1
Heat Number/Lot No.	LT7499EY11
Piece I.D.	1 HEAD
Alloy	INCONEL alloy 718
Product Form	Round
Product Size	11.0000”
Product Condition	Solution Ann and Aged
Lab Heat Treat	None
Image Number	See Photomicrograph
Magnification	500x-100x
Orientation	Long-Cen
Etchant	7-Acids
Etch Time	Standard

NOTE: The recording of false, fictitious or fraudulent statements or entries on this document may be punishable as a felony under Federal Statute.

Testing & Inspection
A PDI Company
4935 Milwee Street
Houston, TX 77092
Phone (713) 939-0440

Ultrasonic Certification Report

Job Details

Report #
NDE-HOU.17846.TX

PO Number
10983

Customer
Premium Alloys

Test Location
Premium Alloys

Material
718 API

HT No.
LT7499EY11

No. Pieces
1

Customer Info:
11" OD x 173" LG

Specification:
CNT-005 Rev. E

Acceptance Criteria:
CNT-005 Rev. E

Equipment

Unit:
Sonatest D70 SN: I010422

Calibration Due Date:
06/12/2026

Part Condition
Clean

RMS
250

Couplant
Cellulose

Comments:
Lot# 11

Longitudinal

Size
1"

MHz
2.25

SN#
19064

Shear Wave

Size
.5x1"

Deg
45

MHz
2.25

SN#
19065

Longitudinal Standardization
1/4" FBH @ 80% FSH

DB
54

Shear Standardization
3% Notch @ 80% FSH

DB
55

Results

Results

Pieces Acceptable
1

Pieces Rejectable
0

Technician
Grant Rankin

Qualification Date
12/2023

Technician Qualifications
ASNT-ACCP Level II

Date
06/25/2025

Material tested at 50-100 degrees F unless otherwise noted.
Material tested at 30% to 60% relative humidity unless otherwise noted.

Reports apply only to the material tested and/or inspected and do not necessarily represent the quality of apparently similar or identical material. Report/Certificate shall not be reproduced except in full, without written approval by Tejas Testing. ISO/IEC 17025 accredidation by PJLA #86703

Ultrasonic Inspection Report

CACTUS JOB #: N/A **CACTUS PO #:** N/A **WO #:** N/A **DATE:** 06/25/2025

SUPPLIER NAME: Premium Alloys **MATERIAL TYPE:** 718 API

EQUIPMENT DESCRIPTION: 11" OD x 230.625" LG **PART NUMBER/REV:** N/A

MATERIAL SIZE/THICKNESS: 11" OD x 230.625" LG **CUSTOMER PRODUCT IDENTIFIER:** N/A

INSPECTION PROCEDURE

NDT SPECIFICATION CNT-005, Ultrasonic Testing **REVISION** E

Cactus Wellhead NDT Specification contains inspection methodology and acceptance criteria

MODE	☑ SHEAR ☑ LONGITUDINAL	SCANNING METHOD	☑ CONTACT ☑ MANUAL ☐ IMMERSION ☐ AUTOMATIC	PRES. TYPE	☑ A-SCAN ☐ PHASED ARRAY ☐ DIGITAL THICKNESS	AREAS EXAMINED	☑ ALL ACCESSIBLE VOLUME ☐ PRESSURE CONTAINING WELD(S) ☐ CLAD WELD(S) ☐ OTHER (EXPLAIN BELOW)

SURFACE CONDITION: 250 RMS / Cleam **MATERIAL THICKNESS/DIA:** 11" OD x 230.625" LG

UT INSTRUMENT MAKE/MODEL: Sonatest D70 **S/N AND CAL. DUE DATE:** I010422 Due: 06/12/2026

COUPLANT: Cellulose Gum Powder **ANGLE:** 0, 45° **CABLE TYPE/LENGTH:** BNC to BNC 6 FT

TRANSDUCER MANUFACTURER: Technisonic **TRANSDUCER SIZE:** 1", .5" x 1" **TRANSDUCER FREQUENCY:** 2.25 MHz

REFERENCE DB: 54, 55 **RECORD ADDITIONAL DB'S FOR:** **CURVATURE:** _____ **ACOUSTIC DIFFERENCES:** _____ **SCANNING:** (+6) **(TOTAL):** 60, 61

REFERENCE STANDARDS

BLOCK SERIAL NUMBER	OVERALL THICKNESS	METAL TRAVEL DISTANCE	REFLECTOR SIZE	REFLECTOR ORIENTATION
SS-16-0400-1200	4.75", 8.75", 12.75"	4", 8", 12"	1/4"	FBH
11459	10.5"	N/A	3%	EDM Notch

INSPECTION RESULTS

WORK ORDER & SERIAL #	HEAT #	HEAT LOT #	SERIAL # (ORIGINAL)	ACCEPT/REJECT QUANTITIES	NCR # (IF APPLICABLE)
	LT7499EY11	11		1 ACCEPT	

SKETCH AND TECHNIQUE DESCRIPTION (Identify and attach additional drawings/sheets):

MIN. THICKNESS REQUIRED:		MIN. THICKNESS RECORDED:		ACCEPT OR REJECT:	

Print Name: Signature:

INSPECTION PERFORMED BY: Grant Rankin **LEVEL:** II **DATE:** 06/25/2025

INSPECTION EVALUATED BY: Grant Rankin **LEVEL:** II **DATE:** 06/25/2025

CUSTOMER REPRESENTATIVE: _____ **DATE:** _____

Signature of the NDE Technician substantiates training to the NDT Specification and attests to inspections performed in accordance with the stated inspection methodology. Supplier NDE Technician qualifications and current eye examinations are provided and maintained by the company.